

$$(1) \sum_{n=0}^{\infty} \frac{n!}{3^n} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{(n+1)!}{3^{n+1}} \cdot \frac{3^n}{n!} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1) \cdot \cancel{3^n} \cdot \cancel{n!}}{3 \cdot \cancel{3^n} \cdot \cancel{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+1}{3} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{3} = \infty \therefore \text{A série diverge.}$$

$$(3) \sum_{n=0}^{\infty} \frac{3^n}{n!} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} \right| = \lim_{n \rightarrow \infty} \frac{3}{n+1} = 0 \quad 0 < 1 \therefore \text{converge}$$

$$(5) \sum_{n=1}^{\infty} \frac{n}{2^n} \rightarrow \lim_{n \rightarrow \infty} \frac{n+1}{2^{n+1}} = \lim_{n \rightarrow \infty} \frac{n+1}{2 \cdot 2^n} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2} \therefore \text{A série converge.}$$

$$(7) \sum_{n=1}^{\infty} \frac{2^n}{n^2} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}}{(n+1)^2} \cdot \frac{n^2}{2^n} \right| = \lim_{n \rightarrow \infty} \frac{2 \cdot n^2}{(n+1)^2} = 2 \cdot \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 2 \cdot \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2 = 2 \cdot \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)^2 = 2 \therefore \text{Diverge}$$

$$(9) \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{(n+1)!} \cdot \frac{n!}{(-1)^n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 \therefore \text{A série converge}$$

$$(11) \sum_{n=0}^{\infty} \frac{n!}{n \cdot 5^n} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{(n+1)!}{(n+1) \cdot 5^{n+1}} \cdot \frac{n \cdot 5^n}{n!} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{5} = \infty \therefore \text{A série diverge}$$

$$(13) \sum_{n=0}^{\infty} \frac{4^n}{n!} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{4^{n+1}}{(n+1)!} \cdot \frac{n!}{4^n} \right| = \lim_{n \rightarrow \infty} \frac{4}{n+1} = 0 \therefore \text{A série converge.}$$

$$(15) \sum_{n=0}^{\infty} \frac{3^n}{(n+1)^n} \rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{\frac{3^{n+1}}{(n+1)^{n+1}}} = \lim_{n \rightarrow \infty} \frac{3}{n+1} = 0 \quad 0 < 1 \therefore \text{A série converge.}$$

$$(17) \sum_{n=0}^{\infty} \frac{4^n}{5^n + 1} \rightarrow \lim_{n \rightarrow \infty} \frac{4^{n+1}}{5^{n+1} + 1} \cdot \frac{5^n + 1}{4^n} = \lim_{n \rightarrow \infty} \frac{4 \cdot 4^n}{5 \cdot 5^n + 1} \cdot \frac{5^n + 1}{4^n} = 4 \cdot \lim_{n \rightarrow \infty} \frac{4 \cdot 3^n + 4}{5 \cdot 3^n + 1} = \frac{16}{3} \cdot \lim_{n \rightarrow \infty} \frac{3^n + 4}{5^n + 1} = \frac{16}{3} \cdot \lim_{n \rightarrow \infty} \frac{1 + \frac{4}{3^n}}{1 + \frac{5}{3^n}} = \frac{16}{3} \therefore \text{A série diverge}$$

$$(19) \sum_{n=0}^{\infty} \frac{(-1)^{n+1} n!}{(2n+1)!} \rightarrow \lim_{n \rightarrow \infty} \frac{(n+1)!}{(2n+3)!} = \lim_{n \rightarrow \infty} \frac{n+1}{4n^2 + 12n + 6} = 0 \quad 0 < 1 \therefore \text{converge}$$

$$(21) \sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n \rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{n}{2n+1} \right|^n} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2} \quad \frac{1}{2} < 1 \therefore \text{A série converge}$$

$$(23) \sum_{n=1}^{\infty} \frac{(-1)^n}{(\ln n)^n} \rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(-1)^{n+1}}{(\ln n+1)^{n+1}} \right|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{(\ln n+1)^{n+1}}} = \lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0 \quad 0 < 1 \therefore \text{A série converge}$$

$$(25) \sum_{n=1}^{\infty} (2\sqrt[n]{n} + 1)^n \rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{|(2\sqrt[n]{n} + 1)^n|} = \lim_{n \rightarrow \infty} 2\sqrt[n]{n} + 1 = \infty \therefore \text{A série diverge}$$

$$(27) \sum_{n=2}^{\infty} \left(\frac{\ln n}{n} \right)^n \rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{\ln n}{n} \right|^n} = \lim_{n \rightarrow \infty} \frac{\ln n}{n} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \frac{1}{1} = 0 \quad 0 < 1 \therefore \text{A série converge}$$

$$(29) \sum_{n=3}^{\infty} \frac{1}{(\ln n)^n} \rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{1}{(\ln n)^n} \right|} = \lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0 \quad 0 < 1 \therefore \text{A série converge}$$

$$(31) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot 5 \rightarrow \text{Teste para séries alternadas: } \frac{5}{n+1} < \frac{5}{n} \text{ e } \lim_{n \rightarrow \infty} \frac{5}{n} = 0 \quad \text{OB} \quad \lim_{n \rightarrow \infty} \frac{5}{n} = 0 \therefore \text{A série converge.}$$

$$(33) \sum_{n=1}^{\infty} \frac{3}{n \sqrt{n}} = \sum_{n=1}^{\infty} \frac{3}{n^{3/2}} \rightsquigarrow \int_1^{\infty} \frac{3}{x^{3/2}} dx = 3 \int_1^{\infty} x^{-3/2} dx = 3 \cdot \left[\frac{x^{-1/2}}{-1/2} \right]_1^{\infty} = \left[-\frac{6}{\sqrt{x}} \right]_1^{\infty} = \left[-\frac{6}{\infty} - (-6) \right] = 6$$

* Condição OB

* Positivo OB

* Decrescente $\rightarrow \frac{3}{(n+1)^{3/2}} < \frac{3}{n^{3/2}}$ OB

$\therefore$ Como integral da valer finita a série converge

(35) $\sum_{n=1}^{\infty} \frac{2^n}{n+1} \rightarrow \lim_{n \rightarrow \infty} \frac{2^n}{n+1} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{2}{1} = 2 \neq 0 \therefore \text{A série diverge}$

(37) $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{3^{n-2}}{2^n} \rightarrow \frac{3^{n-2}}{2^{n+1}} \leq \frac{3^{n-2}}{2^n} \lim_{n \rightarrow \infty} \frac{3^{n-2}}{2^n} = \lim_{n \rightarrow \infty} \frac{3}{2} \cdot \frac{1}{9} = \frac{1}{9} \lim_{n \rightarrow \infty} \left(\frac{3}{2}\right)^n = \infty$ * Teste para série alternada inaplicável

$\hookrightarrow \sum_{n=1}^{\infty} (-1)^n \cdot \frac{3^n}{2^n} \cdot \frac{1}{9} \rightarrow \sum_{n=1}^{\infty} \frac{1}{9} \cdot \left| -\frac{3}{2} \right|^n$ * Série geométrica $\frac{3}{2} > 1 \therefore \text{A série diverge}$ * Devido ao fator pelo teste do raio também

(39) $\sum_{n=1}^{\infty} \frac{10n+3}{n \cdot 2^n}$ $a_n = \frac{10n+3}{n \cdot 2^n}$ $0 \leq \frac{10n+3}{n \cdot 2^n} \leq \frac{1}{2^n}$ * Comparação de limites: $\lim_{n \rightarrow \infty} \frac{10n+3}{n \cdot 2^n} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{10}{2^n} = 10 \cdot 0 = 0 \therefore \text{A série converge}$

(41) $\sum_{n=1}^{\infty} (-1)^n \cdot \ln\left(\frac{n+2}{n}\right)$ * Teste do raio $\lim_{n \rightarrow \infty} \frac{\ln\left(\frac{n+2}{n}\right)}{\ln\left(\frac{n+2}{n}\right)} = \lim_{n \rightarrow \infty} \ln\left(\frac{n(n+3)}{(n+1)(n+2)}\right) = \lim_{n \rightarrow \infty} \ln\left(\frac{n^2+3n}{n^2+3n+2}\right) \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \ln\left(\frac{2n+3}{2n+5}\right) = 0$ $0 < 1 \therefore \text{A série converge}$
 $\log_e 1 = 0 \rightarrow e^0 = 1$

(43) $\sum_{n=1}^{\infty} \frac{\cos n}{2^n}$ Comparação com $\sum_{n=1}^{\infty} \frac{1}{2^n}$, série geométrica convergente convergente que $|\cos n| \leq 1$ então $\frac{|\cos n|}{2^n} \leq \frac{1}{2^n} \therefore \text{A série converge}$

* Teste do raio

(45) $\sum_{n=1}^{\infty} \frac{n^7}{n!}$ $\rightarrow \lim_{n \rightarrow \infty} \left| \frac{(n+1)^7 \cdot \frac{1}{(n+1)!}}{\frac{n^7}{n!}} \right| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$ $0 < 1 \therefore \text{A série converge}$

(47) $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{5^{n-1}}{n!}$ $\rightarrow \lim_{n \rightarrow \infty} \left| \frac{5^{n-1}}{(n+1)!} \cdot \frac{n!}{5^{n-1}} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$ $0 < 1 \therefore \text{A série converge}$
* Teste do raio

(49) $\sum_{n=1}^{\infty} \frac{(-3)^n}{(2n+1)!} = (-1)^n \cdot \frac{3^n}{(2n+1)!} \rightarrow \lim_{n \rightarrow \infty} \frac{3 \cdot 3^n}{(2n+3)!} = \lim_{n \rightarrow \infty} \frac{3^{n+1}}{(2n+3)(2n+2)(2n+1)!} = 0 \therefore \text{A série converge}$